Legion Go Toolbox

The Legion Go's hardware controls, together in the Steam Quick Access menu.

Power and automatic TDP, fan curve, stick and power lights, charge limit, button remapping, swapped menu buttons and per-controller gyro for emulators.

User manual (PDF) · Features · Requirements · Installation · Using the tabs · Troubleshooting · How it works · Credits

Features

One Decky plugin, six tabs (switch with **LB / RB**). Nothing is changed on your device until you switch a feature on, and every feature that your system cannot do says so, with the reason, instead of pretending.

Tab	What you can do
Home	See every module at a glance, limit charging to 80 %, reset
Power	Pick a power mode or set the TDP in watts, per game if you like. **Auto TDP** finds the lowest power that still holds a frame rate you choose
Fan	Draw a ten-point fan curve, presets, full speed, live temperature, per game and on battery vs charger
Lights	Stick lights: effect, colour, brightness, speed. Power-button light on/off, and off while sleeping
Controls	Swap View/Menu with the Legion buttons, remap Y1 Y2 Y3 M2 M3, switch the touchpad off
Gyro	Gyro in ordinary Steam games (handheld), and each detached controller as its own motion device for Dolphin, Switch emulators and others (DSU / CemuHook)

Saved settings are applied when SteamOS starts, after sleep, and when a controller reconnects, also while the menu is closed.

Requirements

Device	Lenovo **Legion Go** (83E1). Built and tested for this model
System	SteamOS with Decky Loader

What needs what

Feature	Works when
Gyro, stick lights, remapping, touchpad	The controllers are connected
Power light	Always on a Legion Go (three ways to reach it are tried)
Charge limit	SteamOS exposes the battery's *Long Life* mode (current versions do)
TDP and power modes	SteamOS has the Lenovo power-limit driver (current versions do)
Auto TDP	The above, in Gaming Mode (the frame rate comes from gamescope)

Feature	Works when
Swapped menu buttons, gyro in Steam games	InputPlumber is running (it is on SteamOS)
Fan curve	**The `acpi_call` kernel module is installed. SteamOS does not include it**

About the fan curve. The Legion Go only accepts a fan table through a Lenovo firmware call, and Linux can only make that call with the `acpi_call` module. Installing a kernel module on SteamOS means changing the read-only system and repeating that after every system update, so this plugin deliberately does not do it for you. If the module is present (for example installed with Legion Center's *Install Fan Support*, or on Bazzite), the curve editor appears by itself. Without it the fan follows the firmware's own curve, which is safe.

Installation

1. Install Decky Loader.
2. If you have **Legion Go Gyro** installed, uninstall it first (this plugin contains it; its tuning and calibration are carried over automatically). Do the same for other plugins that control the same things (SimpleDeckyTDP, Legion Center, HueSync, DeckyPlumber): two plugins steering the same hardware will fight.
3. Decky → settings → **General → Developer mode** on.
4. **Developer → Install plugin from ZIP file** → `release/legion-go-toolbox.zip`.
5. Open the Quick Access menu → Decky → **Legion Go Toolbox**.

Using the tabs

Power

- **Manage power** hands TDP control to the plugin. Turn off Steam's own *TDP limit* in the Performance menu, otherwise both keep overwriting each other.
- **Power mode**: *Custom* gives you a watt slider; the other entries are the firmware's own modes (Quiet, Balanced, Performance).
- **Auto TDP for a frame rate**: choose a target (for example 60 fps). The plugin measures the game's real frame rate and
  - adds power immediately when the frame rate drops below the target,
  - tries one watt less whenever the target has been held for a few seconds,
  - remembers settings that turned out too low, so it settles instead of bouncing.

  Maximum TDP and *Minimum TDP* are the limits it stays within. It works best with a frame limit (in the game or Steam) equal to the target. In menus and loading screens it simply keeps the last setting.
- **Separate settings per game** stores a profile for the running game.
- **CPU boost** off runs cooler for a small performance cost.

Smooth frame rate (per game)

For console-like smoothness, frames must arrive at even intervals, not just average the right number. While a game runs, choose a cap in **Smooth frame rate**. The plugin then:

- adds MangoHud's limiter with `fps_limit_method=early` to the game's launch options (the Linux counterpart of RTSS's async / front-edge limiter; games without MangoHud's Vulkan layer fall back to DXVK's limiter). Your own launch options are kept, and switching off removes exactly what was added.

- switches the screen to a refresh rate the cap divides evenly while the game runs (60 fps: 60 Hz, 72 fps: 144 Hz), with gamescope's own limiter left off, and hands the refresh rate back when the game closes.
- makes Auto TDP aim at that cap and judge the slowest frames, not only the average, so it never cuts power into stutter.

Only caps that divide the screen's refresh rates are offered (Legion Go: 30, 36, 48, 60, 72, 144). Restart the game after changing it. For the best result turn off Steam's Framerate limit and the game's own limiter and V-Sync for that game. Shader-compilation and loading stutter come from the game and cannot be removed this way.

Fan

Needs `acpi_call` (see above). The firmware keeps control of the fan and of overheating protection; the plugin only changes the table it follows.

- The curve always rises with temperature and never goes below Lenovo's own minimum for the Legion Go (44 % at 10 °C rising to 100 % at 90 °C, the same limit Handheld Daemon enforces). *Silent* is exactly that minimum.
- The firmware throws away *full speed* and the custom curve whenever the power mode changes - whether Steam, another tool or this plugin changes it. The plugin reads both back every 5 seconds (and one second after each of its own power changes) and puts them back. The Fan tab shows how often it had to.
- The firmware uses a custom table only in the *Custom* power mode.
- **Separate settings per game** and **Separate settings on charger** give a game, the charger, or a game on the charger its own fan setting. The plugin uses the most specific one that exists (this game on the charger → this game → all games on the charger → all games) and switches by itself when a game starts or closes and when you plug in or unplug. The controls always edit the case you are in right now; *Saved fan settings* lists them all.

Lights

Stick lights are stored in the controllers. *Off while sleeping* switches the power light off when the device suspends and back on when it wakes.

Controls

- **Swap menu buttons** with two layouts:

Layout	Press	Gets
Steam ↔ Menu, QAM ↔ View (default)	Menu / View	Steam / Quick Access
	Legion L / Legion R	Menu / View
Steam ↔ View, QAM ↔ Menu	View / Menu	Steam / Quick Access
	Legion L / Legion R	View / Menu

  Done with an InputPlumber profile on top of the active one, and restored when you switch it off.

- **Remap back buttons** uses Lenovo's own remapping, stored in the controllers. Steam still sees the back buttons as extra buttons too, so leave them unassigned in Steam Input for games where you use this.

Gyro

In Steam games. Steam only offers gyro for controllers that have one, and InputPlumber on its own presents the Legion Go as an Xbox controller. This plugin presents it as a **Steam Deck controller** by default (also selectable: DualSense, DualSense Edge, or back to Xbox), so Steam shows its normal gyro options for every game (Steam button → Controller settings → Gyro).

Choose a type for **all games**, and optionally a different one for the game that is running: it is remembered, switched to when that game starts and back when it closes. *Games with their own type* lists them; select one to remove it.

Button symbols: most games show Xbox symbols whatever type is picked, because Steam Input hands them an Xbox-style controller. Games with their own PlayStation support may show PlayStation symbols with a DualSense (most offer a button-prompt setting); Steam's own menus always match the type. The Steam Deck type keeps A/B/X/Y lettering.

This uses the motion sensor inside the tablet, so it works with the controllers attached; it cannot use a detached controller's own sensor. Switching the type reconnects the controller once, so do it before starting a game. Controller layouts are stored per controller type, so a game may start with that type's default layout.

In emulators. Add a DSU / CemuHook motion server in your emulator: `127.0.0.1`, port `26760`. Slot 0 = whole handheld, slot 1 = left controller, slot 2 = right controller.

- **Dolphin**: *Controllers → Alternate Input Sources → DSU Client*, then in the Wii Remote's *Motion Input* tab choose `DSUClient/2/…` and press *Default*.
- **Ryujinx**: *Input → Motion → Use CemuHook compatible motion*, slot 1 and *Right JoyCon Slot* 2.
- **Eden / Citron / yuzu forks**: *Controls → Advanced → Motion/Touch → Configure*, add the server, then map *Motion 1* / *Motion 2* by shaking a controller.

Press **Calibrate** once and leave the controllers alone for a few seconds.

Troubleshooting

- **A feature says "not available"** – the text under it gives the reason.
- **TDP jumps back** – Steam's own TDP limit is still on, or another TDP plugin is installed.
- **Auto TDP says "waiting for the game to draw frames"** – no game is in focus, or the game is paused.
- **Auto TDP stays at the maximum** – the target frame rate is not reachable within your maximum TDP; raise the maximum or lower the target.
- **Stick lights or remapping report "controllers not found"** – wake the controllers by pressing a button.
- **Menu buttons swap back** – InputPlumber restarted; the plugin re-applies the swap within a few seconds.
- Logs: `~/homebrew/logs/Legion Go Toolbox/`.

Scope and limits

- Used on a Legion Go (83E1), and verified with simulated hardware (137 automated tests). Every module reads the device state back after a change and reports it when a change did not take.
- Legion Go 2 and Legion Go S are recognised for the parts that are known to be identical (power limits, charge limit, power light); the rest stays off.
- Gyro in Steam games comes from the tablet's sensor through SteamOS's own input layer; per-controller gyro is for emulators only.

How it works

```
main.py                      Decky entry point
py_modules/lgo_gyro/         gyro: controller HID reader + DSU server
py_modules/lgo_toolbox/
  hub.py                     applies saved settings, re-applies after sleep
  power.py                   firmware power limits, Auto TDP
  fps_probe.py               game frame rate from gamescope
  fan.py  acpi.py            fan table through Lenovo's firmware call
  powerled.py                power-button light (EC)
  battery.py                 charge limit
  controller.py              stick lights, remapping, touchpad
  plumber.py                 swapped menu buttons (InputPlumber)
src/                         the tabs (React)
tests/                       backend tests with simulated hardware
```

The backend is plain Python (standard library only) and runs as root, as Decky plugins with hardware access do.

Development

```
pnpm install
python3 -m unittest discover -s tests     # backend tests
pnpm run build                            # UI → dist/
bash scripts/package.sh                   # → release/legion-go-toolbox.zip
```

The PDF manual and README live in `docs/`. To rebuild them after editing `docs/src/manual.html` or this README (needs Node with `marked` and Playwright):

```
cd docs/src
node readme.cjs ../../README.md readme.html print.css
node render.cjs && mv *.pdf ..
```

Credits

This plugin is its own code, but it stands on what these projects worked out about the hardware:

- Handheld Daemon and hwinfo – controller protocol, Lenovo firmware methods
- Legion Center / LegionGoRemapper – remapping, fan table, charge limit
- HueSync – power light locations
- SimpleDeckyTDP – Lenovo power-limit interface
- DeckyPlumber and InputPlumber – menu-button swap
- Panel de Control – the idea of reading frame times from gamescope's control interface
- decky-lsfg-vk – tabs in the Quick Access menu
- Legion Go 2 Companion – the module-overview layout

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